Stability Assessment

Nexo InOuty frontend: how stable the system is today, and what fixing the bugs would change

24 October 2026 · companion to *Bug-Report.pdf* (97 bugs, details and effects)

5/10
Today

7.5–8/10
After the **13 most important** fixes

8.5–9/10
After **all 97** bugs are fixed

Verdict

The system runs reliably day to day and nothing crashes in normal use. But the parts that matter most for a payroll system (recorded hours, overtime and pay) give **wrong results in situations the business meets every week**: night shifts, employees working two shifts in one day, and corrections made after the fact. Several of these errors change stored data silently, which makes them hard to notice and costly to correct later.

Score by area

Area	Score	Why
Sign-in, roles, sessions	**8**	Works well; token renewal and page access are reliable. Two security edge cases need a crafted link or look-alike server (BUG-007, BUG-605).
Planning	**6**	Works; one serious race when saving while a day loads (BUG-302) and duplicates from file imports (BUG-305).
Reporting and payroll screens	**5**	Pay formulas are correct, but slow answers can show the wrong period, some employees can be left out, and a negative advance raises net pay (BUG-503).
Clock-in and history	**3**	Most High bugs are here: night shifts, overtime, overwritten records, edits shifted by one hour. These feed directly into payroll.

What the score is based on

Evidence	Result
Automated tests	1,359 passing unit and component tests across 80 files; **98.9%** of lines and 94.2% of branches covered
Known bugs	**97** bugs, each proven by a test that fails until it is fixed: **11 High**, **43 Medium**, **43 Low**
End-to-end tests	12 tests in a real browser on the production build: sign-in per role, page access, session renewal, kiosk, monthly payroll, saving a planning, Arabic
Tablet checks	All pages fit tablet screens; every text field (about 70) brings up the on-screen keyboard; two tablet blockers were fixed

What earns points

- Main workflows work end to end in a real browser.
- Sign-in, automatic session renewal and role-based page access are solid.
- Pay formulas (hourly rate, weekly advance, net pay) match hand calculations.
- Usable on tablets: layouts fit and the keyboard appears wherever typing is needed.
- High test coverage: future changes that break something will be caught.

What costs points

- **Wrong pay in common cases**: night-shift punches paired wrongly, overtime lost or inflated (~21 h), breaks paid as work, a negative advance raising net pay.
- **Silent data loss**: a second shift overwrites the first, adding a punch wipes penalties, saving during loading overwrites a day's planning.
- **Timing and time zones**: slow connections or quick clicks show the wrong period or record twice; punches near midnight land on the wrong day.
- **Hidden errors**: failed loads say "No shifts found"; a failed save can still show "Saved".

How to raise the score

Step	Bugs to fix	Expected score
1. Stop wrong pay and data loss	The 11 High bugs: BUG-015, 101, 102, 206, 209, 217, 218, 221, 302, 406, 503; plus the request races BUG-501 and BUG-504	**about 7.5–8**
2. Fix the rest	The 43 Medium and 43 Low bugs: time zones, error messages, accessibility, display details	**about 8.5–9**
3. Confirm in real conditions	Check the backend behaviours the report flags as "needs backend confirmation" (e.g. BUG-015, BUG-012) and try the system on a real tablet and real clocking machine	**needed for 9+**

Limits of this assessment

The rating covers the **frontend** (the web application). It is based on automated tests run against a simulated backend, not the live server, and on tablet emulation rather than a physical device. Some bugs depend on how the real backend behaves, so a few may prove more or less serious once that is checked. Every bug listed has a test, so each fix can be confirmed objectively: when a bug is fixed, its test starts passing.